FoodFlow: Complete System Architecture

Comprehensive Technical Specification of Dynamic Pricing, Dual-OTP Custody Handover, Real-Time Dispatch Engine, and Courier Escrow Settlement

System Overview: FoodFlow connects Food Surplus Providers (restaurants, caterers, hotels), Food Receivers (NGOs, shelters, relief centers), and Volunteer Couriers. The platform operates on a **Pure Compute-on-Read** dynamic pricing paradigm, a **Dual-OTP cryptographic custody protocol**, and an **Instant Courier Escrow Payout** subsystem, delivering zero data collision across multi-origin local test instances.

1. Multi-Origin Client Architecture & Testing Topology

To allow developers and reviewers to test all 3 actor roles simultaneously in real-time without browser session, cookie, or localStorage collisions, FoodFlow is deployed across 3 dedicated localhost origins. Web browsers isolate local storage and authentication state strictly by origin (port number):

Role / Actor	Port	Direct URL	Key Functional Scope
■■■ **Food Provider**	**8085**	http://localhost:8085/?role=provider	List surplus batches, set base/floor pricing & velocity, simulate real-time price drops, withdraw active listings.
■ **Food Receiver (NGO)**	**8086**	http://localhost:8086/?role=receiver	Browse live rescue catalog, observe 1-second dynamic price decay, view lot valuation savings, claim with instant price-lock guarantee.
■ **Volunteer Squad**	**8087**	http://localhost:8087/?role=volunteer	Discover dispatches, OpenStreetMap live navigation, verify Provider & Receiver Dual-OTPs, view itemized earnings & payment receipts.

2. Mathematical Dynamic Pricing & Markdown Engine

Dynamic pricing uses an **exponential time-decay markdown function**. Prices are calculated in-memory on demand (compute-on-read), eliminating periodic database update cronjobs or repetitive table writes.

$$P_{current} = \max(P_{floor} , P_{base} \times W_{quality} \times W_{quantity} \times (T_{remaining} / T_{total})^{\gamma} \times W_{kitchen})$$

Multiplier / Variable	Notation	Value Matrix	System Behavioral Impact
Base & Floor Bounds	P_{base}, P_{floor}	Configured by Provider	Price will strictly never decay below P_{floor} as long as the food is safe.
Decay Velocity	γ (Gamma)	Staple: 0.5 Normal: 1.0 Perishable: 2.0	Staples hold value longer (concave curve); perishables discount aggressively (convex curve) to force rapid rescue.

Quality Multiplier	$W_{quality}$	Tier 1: 1.0× Tier 2: 0.7× Tier 3: 0.5×	Discounts batches based on initial prep freshness and holding stability.
Volume Multiplier	$W_{quantity}$	<20: 1.0× 20-49: 0.85× ≥50: 0.70×	Bulk incentives: 15% discount for 20+ meals, 30% discount for commercial rescue volume.
Clear the Kitchen	$W_{kitchen}$	0.50× (50% OFF)	Flash markdown override applied instantly when a provider closes commercial shifts.
Strict Expiry Protocol	$T_{rem} \le 0$	$P_{current}$ = ■0.00	Food safety lock: Unsafe batches collapse to ■0.00, marked Expired, and blocked from claims.
Price-Lock Guarantee	Claim Event	lockedUnitPrice lockedTotalPrice	Permanent immutability: The dynamic rate at the millisecond of claim is sealed for the order.

3. Dual-OTP Cryptographic Handover Protocol

To guarantee chain-of-custody verification between commercial kitchens, transit couriers, and beneficiary NGOs, FoodFlow enforces a two-stage OTP authentication workflow:

Milestone	Actor Required	Verification Token	Security & System Action
Stage 1: Provider Pickup	■■■ Provider → ■ Courier	**Pickup OTP** (e.g. 4821)	Entered by volunteer at provider dock. Validates physical meal handover and food temperature checks. Transitions status to pickedUp.
Stage 2: Receiver Delivery	■ NGO → ■ Courier	**Delivery OTP** (e.g. 7390)	Disclosed by NGO intake coordinator upon receipt. Proves safe delivery and immediately seals order to delivered, triggering escrow disbursal.

4. Batch Lifecycle State Machine

Surplus batches progress through a deterministic, crash-resilient state machine managed by FoodRescueService:

Status Code	Triggering Event	Actor	System State Transitions & Side Effects
available	listNewBatch()	Provider	Generates Pickup & Delivery OTPs. 1-second live pricing decay starts.
claimed	claimBatch()	Receiver (NGO)	Locks unit & total price. Notifies volunteer network for courier dispatch.
inTransitToPickup	acceptTask()	Volunteer	Assigns courier ID. Starts OpenStreetMap GPS pickup routing.
pickedUp	verifyPickupOtp()	Volunteer + Provider	Validates pickup OTP match. Re-routes GPS navigation to receiver dropoff dock.
delivered	verifyDeliveryOtp()	Volunteer + Receiver	Validates delivery OTP match. Closes rescue, generates banking UTR, triggers instant UPI.
withdrawn	withdrawListing()	Provider	Provider revokes available listing with confirmation dialog and toast alert.
expired	$T_{remaining} \le 0$	Automated Ticker	Auto-flips when safeUntil passes. Price resets to ■0; batch locked from claims.

5. Volunteer Payments & Escrow Confirmation System

Upon delivery OTP validation, the courier receives an instant transparent payout via FoodFlow Smart Courier Escrow. Couriers retain 100% of their compensation with complete banking auditability:

Compensation Line Item	Standard Rate	Sample Calculation (4.2 km, 50 meals)	Purpose / Incentive
Base Transit Allowance	■50.00 flat	■50.00	Guarantees courier base pay for dispatch commitment.
Distance Compensation	■12.00 / km	■50.40 (4.2 km)	Fair fuel and transit reimbursement calculated via GPS routing.
Safe-Window Rapid Bonus	■30.00 bonus	■30.00	Rewarded for arriving within thermal safety threshold.
Bulk Lot Handling Incentive	■30.00 (≥40 meals)	■30.00	Compensates courier for heavy load management.
Total Courier Payout	**100% Retained**	**■160.40**	**Disbursed instantly to UPI VPA (volunteer.courier@okaxis)**

Audit & Confirmation Architecture:

• **VolunteerPaymentConfirmationPage:** Success hero badge (PAID ✓), dual-OTP signatures, dynamic lot settlement, and banking UTR.

• **VolunteerPaymentsPage:** 4 metrics (Lifetime Earned ■3,840, 24 Rescues, Avg ■160, 100% Settled), UPI VPA editor, and receipt history.

• **Workflow Integration:** Immediate post-delivery prompt, history item direct actions, and home screen quick-access banner.

• **Official PDF Receipt:** Tax-exempt delivery certificate generated with immutable reference ID (e.g. TXN-FF-TESTPAY1).

6. Verification, Automated Testing & Code Quality

The implementation maintains comprehensive test coverage and clean static analysis across all layers of the codebase:

Test Suite File	Target Domain	Passing Tests	Key Invariants Verified
volunteer_payment_confirmation_test.dart	Volunteer Payments	**3 / 3**	Earnings breakdown, Dual-OTP badges, banking UTR audit, UPI management, responsive layout.
dynamic_pricing_service_test.dart	Dynamic Pricing & Decay	**14 / 14**	Continuous time-decay, gamma velocities (0.5, 1.0, 2.0), quality tiers, volume discounts, floor price clamp, price locking.
food_rescue_service_test.dart	Lifecycle & Dual-OTP	**5 / 5**	Pickup & Delivery OTP generation, handover validation, courier dispatch, Supabase serialization.
provider_withdraw_test.dart	Provider Management	**2 / 2**	Listing withdrawal confirmation, dialog cancel behavior, immediate UI removal.

widget_test.dart & Integrations	Navigation & Auth	**38 / 38**	End-to-end multi-role routing, signup validations, map embedding, adaptive desktop/mobile views.
Total Project Test Coverage	**All Subsystems**	**62 / 62 PASS**	**100% Pass Rate · 0 Failures · 0 Flaky Tests**

Static Code Analysis: flutter analyze passed cleanly with **0 errors, 0 warnings, 0 lints**.